SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Vishal.V
192419033

Programme: **B.E / B.Tech**

Course Code: **CSA1741**

Course Name: **ARTIFICIAL INTELLIGENCE**

Duration: **1: 30 Hr**

Date: **02-11-2026**

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.
CO2	Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently.

Answer all the Questions

1. **SmartSignal – agent design analysis.** A city traffic department installs SmartSignal, an autonomous controller for a busy four-way junction. It reads queue lengths from road-embedded loop detectors and cameras on all four approaches, receives priority requests from ambulances and fire engines, and switches the green, amber and red phases. Camera views are blocked by large buses, drivers may jump the signal, and traffic surges unpredictably when nearby schools and offices close.

   (a) Specify the task environment of SmartSignal using the PEAS description (Performance measure, Environment, Actuators, Sensors). **[2]**

   (b) Characterize the task environment on six properties (fully/partially observable, single/multi-agent, deterministic/stochastic, episodic/sequential, static/dynamic, discrete/continuous) and justify each with evidence from the scenario. **[3]**

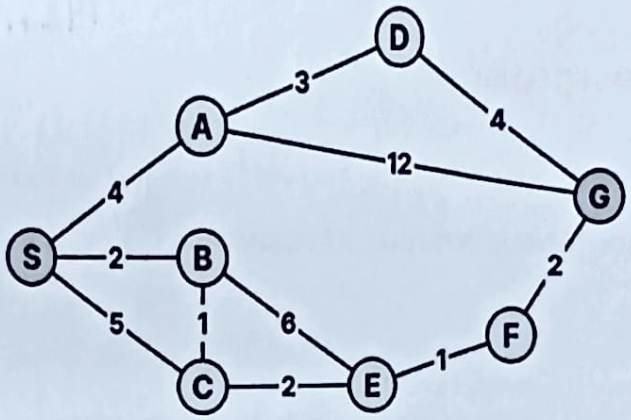
   (c) Two controllers are tested over the same 2-hour evening peak with the performance measure: **+5** per vehicle cleared, **−0.5** per vehicle-minute of waiting, **−50** per emergency vehicle delayed by more than 1 minute and **−20** per safety incident. **Agent A** (fixed-cycle simple reflex): 2,400 vehicles cleared, 9,600 vehicle-minutes of waiting, 3 delayed emergency vehicles, 4 incidents. **Agent B** (model-based, utility-based): 2,750 vehicles cleared, 5,200 vehicle-minutes of waiting, no delayed emergency vehicles, 1 incident. Compute the total score and the average waiting time per vehicle for each agent, and analyze which one is more rational. Explain why an agent can be rational even when an unforeseen surge causes a long queue. **[3]**

   (d) Each of the four approach sensors reports one of 5 queue levels and an emergency flag (present/absent) is received each cycle. Compute the number of distinct percepts per cycle and the

number of entries a table-driven agent would need for a lifetime of only $T = 3$ cycles ($\Sigma|P|^t, t = 1...T$). Analyze why the table-driven design is infeasible and justify the agent structure that should replace it. **[2]**

(10) (CO1) (BL4)

2. **Warehouse picking robot – problem formulation and uninformed search.** An automated warehouse has eight aisle junctions: S (charging dock), A, B, C, D, E, F and G (packing station). Two-way aisles and their travel times in seconds are: S–A 4, S–B 2, S–C 5, A–D 3, A–G 12, B–C 1, B–E 6, C–E 2, D–G 4, E–F 1, F–G 2 (see figure).



(a) Formulate the picking-route task as a single-state problem: state, initial state, goal test, actions, transition model and path cost. State the size of the state space and justify the level of abstraction chosen (for example, ignoring battery level and shelf heights). **[2]**

(b) Trace BFS (goal test when a node is generated), DFS (recursive, neighbours visited in alphabetical order, goal test when a node is visited) and UCS (goal test when a node is selected for expansion). Use graph search with an explored set; break ties alphabetically. For each strategy give the order of nodes expanded, the solution path and its cost. **[4]**

(c) (i) Analyze why BFS and DFS return slower routes than UCS on this map. (ii) The planner is scaled to a mega-warehouse with branching factor $b = 6$, the nearest packing station at depth $d = 10$ and maximum depth $m = 15$. Each node needs 2 KB and the controller generates 5×10^5 nodes/s. Compute the nodes generated by BFS ($b + b^2 + \ldots + b^d$) with its memory and time, the nodes generated by iterative deepening ($d \cdot b + (d-1)b^2 + \ldots + 1 \cdot b^d$) with its time and percentage overhead over BFS, and the memory of DFS ($b \cdot m + 1$ nodes). (iii) Evaluate whether any uninformed strategy meets a limit of 60 s and 4 GB per order, and justify the kind of strategy that is needed. **[4]**

(10) (CO1) (BL4)

3. (i) **Sensorless tray shuttle – search with partial information.** A tray shuttle runs on a straight rail with docks numbered 1 to 4 (left to right). Each dock may hold a jammed tray or be free. After a power failure the shuttle has lost both its position encoder and its jam sensor, so it receives no percepts. Its actions are Left, Right and Clear (Clear frees the jam at the current dock; Left at dock 1 or Right at dock 4 leaves the shuttle where it is). The goal is that all four docks are free.

(a) For a rail with n docks, derive the number of physical states and the number of possible belief states. Compute both for $n = 3$ and $n = 4$. **[1]**

(b) Starting from the initial belief state (complete ignorance, $n = 4$), construct the belief-state sequence for the plan [Left, Left, Left, Clear, Right, Clear, Right, Clear, Right, Clear] and state the size of the belief state after every action. **[2]**

(c) Analyze why the three initial Left actions are needed (localization by coercion), and evaluate how the plan and search effort change if (i) the Left action is slippery and may fail to move the shuttle, and (ii) the jam sensor is repaired so that the problem becomes a contingency problem. **[2]**

(5) (CO1) (BL4)

(ii) **Campus Wi-Fi channel allocation – constraint satisfaction.** Five access points P1–P5 in a library block must each be assigned one of the non-overlapping channels {1, 6, 11}. Access points whose coverage overlaps must use different channels: P1–P2, P1–P3, P2–P3, P2–P4, P3–P4, P3–P5 and P4–P5. In addition, P1 is a legacy unit fixed to channel 1, and P4 cannot use channel 11 because of interference from a nearby radar.

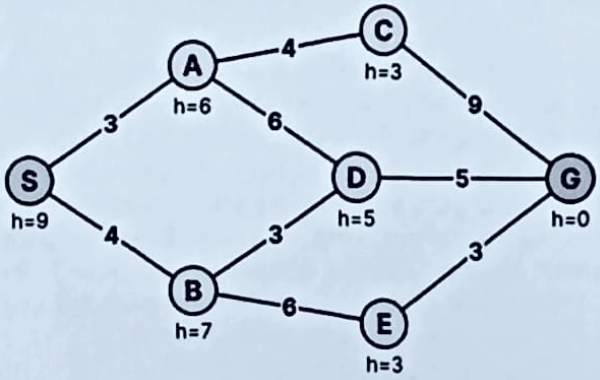
(a) Formulate the problem as a CSP (variables, domains, constraints) and draw its constraint graph. **[1]**

(b) Apply node consistency and then solve by backtracking search with the MRV heuristic (ties: degree heuristic, then lowest index; values tried in ascending order) and forward checking. Show the domains of all variables after every assignment and report the number of backtracks. **[3]**

(c) Determine the total number of valid channel plans, and evaluate how node consistency, MRV and forward checking reduced the search compared with the unconstrained space of $3^5 = 243$ assignments. **[1]**

(5) (CO2) (BL5)

4. **Medicine-delivery drone – informed search and heuristics.** A drone must fly from the depot S to the rural health centre G over the two-way air corridors shown (flight time in minutes): S–A 3, S–B 4, A–C 4, A–D 6, B–D 3, B–E 6, C–G 9, D–G 5, E–G 3. The estimated flight time to G, supplied by the navigation module, is h(S)=9, h(A)=6, h(B)=7, h(C)=3, h(D)=5, h(E)=3, h(G)=0.



(a) Run greedy best-first search (graph search). Give the order of expansion, the path and its cost, and analyze why it fails to find the fastest route. **[2]**

(b) Run A* with f(n) = g(n) + h(n). Present a table showing the frontier with g, h and f values at each step; give the order of expansion, the path and its cost. **[3]**

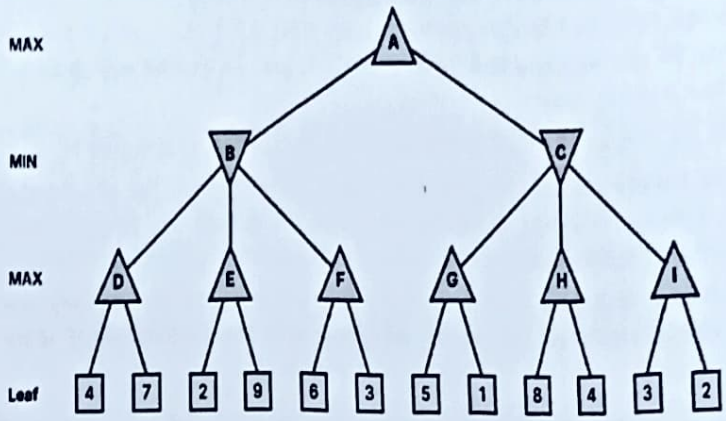
(c) Verify whether h is admissible and consistent by computing the true cost $h^*(n)$ of every node and checking the consistency condition $h(n) \le c(n, n') + h(n')$ on every corridor in both directions. **[2]**

(d) To save on-board computation, the team proposes weighted A* with $f(n) = g(n) + 2 \cdot h(n)$. Re-run the search and evaluate its effect on the number of nodes expanded and on optimality, compared with A*. Suggest how an admissible heuristic for the drone could be derived from a relaxed version of the problem. **[3]**

(10) (CO2) (BL5)

5. **Chess-training engine – adversarial search.** A chess-training engine (MAX) searches three plies ahead against a student (MIN). The game tree is shown below; the utilities of the 12 leaf positions are

given left to right.



(a) Compute the minimax value of every node and identify the best move at the root and the principal variation. **[2]**

(b) Apply alpha-beta pruning (left to right). Show the (α, β) values at every node visited, mark each pruned branch and the cut-off that caused it, and count the leaves evaluated compared with plain minimax. Reorder the children of B and C to give the best-case ordering and state the leaves evaluated. **[4]**

(c) For a game with branching factor b = 30 and a search depth of d = 8 plies, compute the leaf positions examined by minimax and by alpha-beta with best-case move ordering (≈ b^(d/2)), and the time of each at 10^9 positions per second. Estimate the depth each can reach in 1 s. **[2]**

(d) The engine uses Eval(s) = 1·(P_MAX − P_MIN) + 3·(N_MAX − N_MIN) + 3·(B_MAX − B_MIN) + 5·(R_MAX − R_MIN) + 9·(Q_MAX − Q_MIN) + 0.1·(mobility_MAX − mobility_MIN), where P, N, B, R, Q are the numbers of pawns, knights, bishops, rooks and queens. Evaluate a position where MAX has 6 pawns, 1 knight, 2 bishops, 2 rooks, 1 queen and 32 legal moves, while MIN has 7 pawns, 2 knights, 1 bishop, 2 rooks, 1 queen and 27 legal moves. Discuss how a fixed cut-off depth can mislead the engine (horizon effect) and how quiescence search helps. **[2]**

(10) (CO2) (BL5)

Solutions:-

Q1) a) PEAS description.

PEAS describes the working environment of an intelligent agent.

1. Performance: Reduce traffic waiting time, clear more vehicles and avoid accidents
2. Environment: four-way junction, Vehicles, Pedestrians and emergency Vehicles.
3. Actuators: Red, Yellow and Green traffic lights.
4. Sensors: Cameras, loop detectors and emergency request systems.

b) characteristics of the task environment:

1. Partially observable: Cameras may be blocked by buses
2. Multi-agent: Drivers and emergency Vehicles interact with the system.
3. Stochastic: Traffic movement are unpredictable
4. Sequential: Each signal decision affects future traffic

(c) Calculations:

Formula:

$$Score = 5V - 0.5W - 50E - 20I$$

Agent A:

Given:

$V = 2400$, $W = 9600$, $E = 3$, $I = 4$

$$Score = 5(2400) - 0.5(9600) - 50(3) - 20(4)$$

$$= 12000 - 4800 - 150 - 80$$

$$\boxed{Score = 6970}$$

Average waiting time:

$$= \frac{9600}{2400} = 4 \text{ min}$$

Agent B:

Given:

$V = 2750$, $W = 5200$, $E = 0$, $I = 1$

$$Score = 5(2750) - 0.5(5200) - 50(0) - 20(1)$$

$$= 13750 - 2600 - 20$$

$$\Rightarrow \boxed{Score = 11130}$$

Average waiting time:

$$\frac{5200}{2750} = 1.89 \text{ min}$$

Conclusion:- Agent B is more rational because it clears more vehicles, reduces waiting time, avoids emergency delays and has fewer safety

(a) Calculation:-

No of Percepts:

$|P| = 5^4 \times 2$

$= 625 \times 2$

1250 Percepts

Table entries for T=3

$P + P^2 + P^3$

$= 1250 + (1250)^2 + (1250)^3$

$= 1250 + 1562500 + 1953125000$

$= 1,954,688,750$ entries

Table-driven agent not Suitable because:

- It requires huge memory
- It cannot efficiently handle all Possible Situations
- It is difficult to update When traffic conditions change.

Replacement:- A model-based, utility-based agent is Preferred because it maintains information about the environment and Selects the best action based on expected benefits.

2Q) (a) Problem formulation

1. State: Current Position of the robot (S, A, C, D, E, F, G).
2. Inital state: S (charging dock)
3. Goal test: Reach G (Packing dock)
4. Actions: Move to connected junction
5. Transition model: Movement between connected junctions.
6. Path cost: Total travel time in seconds.

State space:- 8 junctions

(b) BFS, DFS and UCS

1: BFS (Breath-First Search)
- Search level by level.
- Expansion order: S, A.
- Solution Path: S → A → G
Cost:
4 + 12 = 16 sec

2: DFS (Depth-First Search)
- Explores one path deeply before returning
- Visit order: S → A → D → G
- Solution Path: S → A → D → G
Cost:
4 + 3 + 4 = 11 sec

3. UCS (Uniform Cost Search)

- Search the node and Select the node with the lowest total Path Cost.
- Expansion order: S → B → C → A → E → F → D → G.
- Solution Path: S → B → C → E → F → G.

Cost:

2 + 1 + 2 + 1 + 2 = 8 Sec

UCS finds the cheapest route because it considers the actual travel cost.

(c) Calculations :-

Why are BFS and DFS slower than UCS?

- BFS considers the number of edges not the travel time.
- DFS follows a deep Path without considering total cost.

Mega warehouse Calculation :-

Given:-

- $b = 6$
- $d = 10$
- $m = 15$
- Memory Per node = 2KB
- Speed = 500,000 nodes/Second

1. BFS

$N = b + b^2 + \dots b^0$

$N = 72,559,410$

Memory:

$72,559,410 \times 2$

$= 145,118,820$ KB

Time:

$\frac{72,559,410}{500000} = 145.12$ Sec

2. IDS (Iterative Deepening Search)

$N = db + (d-1)b^2 + \dots + b^d$

$N = 87,071,280$

Time: $\frac{87,071,280}{500000}$

$= 174.14$ Sec

Overhead:

$\frac{IDS - BFS}{BFS} \times 100$

$= 20\%$

3. DFS Memory:

$= b \times m + 1$

$= b \times 15 + 1$

91 nodes = 182 KB

- BFS exceeds 60 Sec and 4GB
- IDS exceeds 60 Sec
- DFS uses less memory but cannot guarantee an optimal solution.

Conclusion: A* Search is Preferred, because it uses heuristic information searching.

(i) Sensorless Taxy Shuttle:

Q3) (a) Physical and belief states:

Each dock can be free or jammed.

Formulas:

$$Physical = n \times 2^n$$

$$Belief = 2^{n \times 2^n}$$

For $n = 3$

$$Physical = 3 \times 8 = 24$$

$$Belief = 2^{24} = 16,777,216$$

For $n = 4$

$$Physical = 4 \times 16 = 64$$

$$Belief = 2^{64} = 18,446,744,073,709,551,616$$

(b) Belief-State Sequence:

Given Plan:- Left, Left, Left, Clean, Right, Clean, Right, Clean, Right, Clean.

initially: $4 \times 16 = 64$

Sequence:-

- initial = 64
- Left = 48
- Left = 32
- Left = 16
- Clean = 8
- Right = 8
- Clean = 4
- Right = 4
- Clean = 2
- Right = 2
- Clean = 1

Three Left actions [illegible]
Then it clears each dock while moving right.

Result: All four docks become free.

(c) Changes in Plan:

(i) Slippery Left action:

- The shuttle may fail to move
- Its position becomes uncertain
- more actions may be required

(ii) Jam Sensor Repaired:

- The shuttle can identify jammed docks.
- it can make decisions using sensor information
- A contingency plan can replace the fixed plan.

(ii) Campus WiFi - channel Allocation:

(a) CSP formulation:

Variables: P_1, P_2, P_3, P_4, P_5

Domains:

$$\{1, 6, 11\}$$

Constraints:

- $P_1 \neq P_2$, $P_1 \neq P_3$
- $P_2 \neq P_3$, $P_2 \neq P_4$
- $P_3 \neq P_4$, $P_3 \neq P_5$
- $P_4 \neq P_5$

Additional condition:

$$P_1 = 1, \quad P_4 \neq 11$$

(b) Node Consistency and MRV

Node Consistency:

- $P_1 = 1$
- $P_4 = 6$
- $P_2, P_3, P_5 = 11$

Apply MRV and forward checking

MRV selects the Variable with the fewest available channel 1.

Select $P_3 = 6$.

Forward checking gives:

- $P_2 = 11$
- $P_4 = 1$
- $P_5 = 11$

Then assign:

- $P_2 = 11$
- $P_4 = 1$
- $P_5 = 11$

So now:-

$P_1 = 1, P_2 = 11, P_3 = 6, P_4 = 1, P_5 = 11$

No of backtracks in each search is 0

(c) Total Valid channel Plan

$3^5 = 243$

initially there are 243 Possible assignments.

After applying all constraints:

2 Valid channel Planes

Node Consistency, MRV and forward checki

by eliminating

Q4) (a) Greedy Best-First Search

Greedy Search Selects the node with the lowest heuristic Value(h).

Steps:

1. start at S.
2. Select A (h=6)
3. Select C (h=3).
4. Select G (h=0).

Expansion order: S → A → C → G.

Path: S → A → C → G.

Cost:

3+4+9 = 16 min

Greedy Search does not find the fastest route because it ignores the actual travel cost.

(b) A* Search

A* Search Selects the node with the lowest total cost.

formula:

$f(n) = g(n) + h(n)$

- $g(n)$ = Actual cost.
- $h(n)$ = Estimated cost.

Expansion order: S → A → C → B → D → G

Path: S → B → D → G

Cost: 4+3+5 = 12 min

(c) Admissibility and Consistency

Heuristic never overestimates the actual cost.

Heuristic Satisfies:

$$h(n) \leq c(n, n') + h(n')$$

Actual Shortest Costs to G_1:

- $S = 12, A = 11, B = 8$
- $C = 9, D = 5, G = 0$

All nodes satisfy both conditions.

The heuristic is admissible and consistent.

(d) Weighted A* Search

formula $f(n) = (g(n) + 2h(n))$

1. Start at S
2. Select A (f = 15)
3. Select C (f = 13)
4. Select G_1 (f = 16)
5. Goal reached

Expansion order: $S \Rightarrow A \Rightarrow C \Rightarrow G_1$

Path: $S \Rightarrow A \Rightarrow C \Rightarrow G_1$

Cost: 16 min

Q5) (a) Minimax Values

- Max Selects the highest Value
- Min Selects the lowest Value

Max nodes:

- D = max(4,7) = 7
- E = max(2,9) = 9
- F = max(6,3) = 6
- G = max(5,1) = 5
- H = max(8,4) = 8
- I = max(3,2) = 3

Min nodes:

B = min(7,9,6) = 6

C = min(5,8,3) = 3

Root:

A = max(6,3) = 6

Best move: B

Principle Variation: A → B → F → 6

(b) Alpa-Beta Pruning

Alpha (α): Best Value for Max

Beta (β): Best Value for Min

Pruning Condition: $\alpha \geq \beta$

Steps:-

1. Evaluate B: D=7, E=9, F=6. Therefore B=6
2. At root A, $\alpha = 6$.
3. Evaluate G=5 under C. Therefore B=5
4. Since $\alpha \geq \beta$, Prune H and I

Result:-

- Minimax evaluates 12 leaves
- Alpha-Beta evaluates 8 leaves.
- 4 leaves are Pruned.

Best-Case ordering:

- B: F → D → E
- C: I → G → H

(c) Calculations:-

Given: b = 30, d = 8, Speed = 10^6 Positions/Second

Minimax:

$$N = 30^8$$

$$\Rightarrow 656,100,000,000$$

Time

$$= \frac{656,100,000,000}{10^6}$$

$$= 656,100 \text{ Sec}$$

Alpha-Beta (best case):

$$N = 30^{8/2} = 30^4$$

$$= 810,000$$

Time:-

$$= \frac{810,000}{10^6} = 0.81 \text{ Sec}$$

Depth reachable in 1 Sec:

- Minimax = 4 Piles
- Alpha-Beta = 8 Piles

(d) Function and Horizon Effect:-

$$Eval = 1(6-7) + 3(1-2) + 3(2-1) + 5(2-2) + 9(1-1) + 0.1(32-27)$$

$$= -1 - 3 + 3 + 0 + 0 + 0.5$$

$$\boxed{Eval = -0.5}$$

The position is slightly favourable to Min

Horizon effect:
Occurs when the search stops at a fixed depth and misses an important future move, such as losing a queen.

Quiescence Search:
Extends the search in unstable positions by examining captures and checks to improve the evaluations.